

3.1 Baseline Drift

The detector baseline drifts upward when the nebulizer gas pressure falls below the specified range. Confirm the regulator setting, then inspect the supply line for a partial blockage before replacing the detector cell. A slow drift over several hours usually points to temperature rather than gas supply.

3.2 High Backpressure

Replace the inlet frit when backpressure rises above the documented limit for the installed column. Flush the system with the recommended solvent for twenty minutes and record the resulting pressure. If pressure does not fall, isolate the pump head and repeat the measurement with the column removed.

3.3 Retention Time Shift

Retention times shift earlier when the mobile phase composition changes. Verify the proportioning valve seal and check the solvent reservoir levels. Degas the mobile phase before every sequence, because dissolved air produces irregular flow and irreproducible retention across the run.